SmartEduLoan

Product Requirements Document (PRD)

A Full-Stack Student Loan Management System

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1. Product Overview

SmartEduLoan is a web-based platform designed to simplify and automate the complete lifecycle of student loan management. It brings together three key stakeholders — **Students**, **Administrators**, and **Banks** — on a single, unified digital platform, replacing the traditional paper-based and fragmented loan processes with a streamlined, transparent, and automated system.

The platform enables students to apply for education loans online, tracks their application through review stages, and provides real-time status updates. Admins manage the review pipeline and route applications to partner banks, while banks perform verification through their own dedicated interface.

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2. Problem Statement

Student loan management in educational institutions is often slow, opaque, and manually intensive:

- Students have **no visibility** into where their application stands after submission.
- Administrative staff spend excessive time **manually reviewing and routing** applications.
- Banks lack a **structured digital channel** to receive, review, and process assigned loan requests.
- There is **no automated mechanism** to pre-screen ineligible applicants, leading to wasted time on both sides.
- Loan repayment tracking (EMIs) is disconnected from the application system.

SmartEduLoan addresses all of these problems by creating an end-to-end automated workflow with real-time communication between all stakeholders.

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3. Goals & Objectives

Goal	Description
Digitize loan application	Allow students to submit loan applications with documents entirely online
Automate credit evaluation	Compute credit scores from student profiles without manual input

Streamline admin review	Give admins a single dashboard to track, assign, approve, or reject applications
Enable bank verification	Provide banks a dedicated portal to receive and review assigned applications
Improve transparency	Let students view exactly where their loan is in the pipeline at any time
Automate repayment planning	Generate and display EMI schedules automatically upon loan approval

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4. Target Users

Role	Who They Are	What They Do
Student	Enrolled college/university students seeking education financing	Apply for loans, upload documents, track status, view EMI schedule
Admin	Institution administrators who oversee the loan process	Review applications, assign them to banks, approve or reject
Bank	Registered financial institutions partnered with the institution	Verify student documents and submit verdicts on assigned applications

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5. Key Features

5.1 Authentication & Role-Based Access

Students, admins, and banks each register and log in with separate credentials. The system issues a **JWT token** upon login, which controls access to role-specific portals. No user can access another role's data or functionality.

5.2 Student Portal

- **Profile Setup:** Students fill in academic details (institution, course, year, percentage), financial background (income, existing loans), and address information.
- **Loan Application:** Students apply for a loan by specifying the requested amount, tenure, purpose, and uploading documents.
- **Status Tracking:** A real-time timeline shows every stage of the application — submitted, under review, assigned to bank, approved, or rejected.
- **Credit Score Display:** Students can see their computed credit score on their profile page.

5.3 Admin Portal

- **Dashboard Overview:** Summary cards showing total, pending, approved, and rejected application counts.
- **Application Management:** Searchable and filterable list of all loan applications with quick-action buttons.

- **Bank Assignment:** Admins can route any application to a registered bank for verification.
- **Approve / Reject:** Final decision-making with a mandatory reason field on rejection.
- **Bank Management:** Admins can add new banks and view existing registered bank partners.

5.4 Bank Portal

- **Assigned Applications:** Banks only see the loan applications that have been routed to them by the admin — no cross-bank data leakage.
- **Document Review:** Banks can view all uploaded student documents and profile data for due diligence.
- **Verification Verdict:** Banks submit their verification outcome, which feeds back to the admin for final decision.

5.5 Automated Credit Scoring

The system automatically calculates a numeric credit score for each student based on their profile data — academic performance, income-to-expense ratio, and existing liabilities. **Any loan application where the credit score falls below 600 is automatically rejected** before it reaches the admin, immediately notifying the student with a reason.

5.6 Loan Product Catalog

Banks can define and manage their own loan products within the system, specifying applicable interest rates, minimum/maximum tenure, and maximum loan amount. Students can browse eligible products when applying.

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6. User Stories

ID	As a...	I want to...	So that...
US-01	Student	Register and log in securely	I can access my personal loan portal
US-02	Student	Fill out my academic and financial profile	The system can evaluate my creditworthiness
US-03	Student	Apply for a loan and upload required documents	I can formally request education financing
US-04	Student	Track my loan status in real time	I know exactly where my application stands
US-05	Student	See my loan status and bank details after approval	I can track progress after approval
US-06	Admin	View all pending loan applications in one place	I can process them efficiently without missing any

US-07	Admin	Assign an application to a bank	The bank can independently verify the student's case
US-08	Admin	Approve or reject an application with a reason	Final decisions are recorded and communicated clearly
US-09	Bank	View only the applications assigned to me	I focus on my workload without seeing unrelated cases
US-10	Bank	Review student documents before submitting a verdict	My verification is well-informed and accurate
US-11	System	Automatically reject applications with credit score < 600	Only eligible applicants reach the human review stage

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7. Non-Functional Requirements

Category	Requirement
Performance	All API responses should be returned within 500ms under normal operating conditions
Security	All routes (except login and registration) must be protected by JWT authentication; passwords must be hashed using bcrypt

Usability	The interface should be intuitive enough for non-technical students and admins to navigate without training
Scalability	The backend must follow a modular architecture (routes, models, services) to allow future expansion
Reliability	File uploads must be validated for correct file type and size before being stored
Maintainability	Code must follow consistent conventions and separation of concerns

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8. Tech Stack

Layer	Technology	Purpose
Frontend	HTML5, CSS3, Vanilla JavaScript	User interface, dynamic content without a framework
Backend	Node.js, Express.js	REST API server, business logic
Database	MongoDB (via Mongoose ODM)	Persistent data storage
Authentication	JSON Web Tokens (JWT)	Stateless, secure session management

File Handling	Multer	Multipart form data and document uploads

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9. Constraints

- **No frontend framework:** The UI is built with pure HTML, CSS, and JavaScript — no React, Angular, or Vue.
- **Local deployment only:** The application runs on localhost during development; no cloud hosting is required.
- **Single database instance:** One MongoDB database serves all roles and data without replication.
- **No live payment gateway:** EMI "Pay Now" is a UI interaction only; no real financial transaction is processed.

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10. Success Metrics

- A student can complete the full loan application flow — registration to submission — without errors or confusion.
- An admin can approve, reject, or assign an application in fewer than 5 interactions.
- The credit score auto-rejection fires correctly for all applications with a score below 600.
- All three portals (Student, Admin, Bank) are cleanly separated, with no unauthorized cross-role data access.